

**Rensselaer Polytechnic Institute**  
**Department of Electrical, Computer, and Systems Engineering**  
**ECSE 2500: Engineering Probability, Spring 2026**

Homework #6: due Thursday, March 19<sup>th</sup>, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.  
While you can check your answers with Wolfram Alpha, remember that you're going to have to do  
integrals by hand on the exam!

1. (30 points.) Alysa and Eileen compete in a snowboarding competition where a Gold medal is worth 3 points, a Silver medal is worth 2 points, and a Bronze medal is worth 1 point (0 points are awarded if the athlete doesn't get a medal). Eileen has a  $\frac{1}{4}$  probability of getting each point value in  $\{0, 1, 2, 3\}$ . Alysa has a  $\frac{2}{3}$  probability chance to beat Eileen (with each of the possible outcomes being equally probable) and a  $\frac{1}{3}$  probability chance to lose to Eileen (with each of the possible outcomes being equally probable). It is not possible for them to tie (e.g., if Eileen gets 3 points, then Alysa must get 2, 1, or 0 points, each with conditional probability  $\frac{1}{3}$ ). Let  $A$  be Alysa's score, and  $E$  be Eileen's score.

- (a) (10 points.) Compute the joint PMF  $p_{AE}(x, y)$ . Represent your answer as a table.
- (b) (10 points.) Compute the joint CDF  $F_{AE}(x, y)$ . Represent your answer as a 2D piecewise constant function (i.e., divide the  $XY$ -plane into grid squares and indicate the height of the CDF at each point).
- (c) (10 points.) Compute the marginal PMF  $p_A(y)$ . Represent your answer as a table.

2. (30 points) Consider the function

$$g(x, y) = \begin{cases} cx(1-x)y & x \in [0, 1], y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (7 points) Determine the value of  $c$  so that  $g(x, y)$  is a valid joint PDF  $f_{XY}(x, y)$  for two random variables  $X$  and  $Y$ .
- (b) (7 points) Compute the joint CDF  $F_{XY}(x, y)$  corresponding to the PDF in part (a). You only need to compute the CDF for values  $(x, y)$  such that  $x \in [0, 1], y \in [0, 1]$ .
- (c) (7 points) Compute the marginal PDF  $f_X(x)$ . Remember to specify the PDF at all values of  $x$ .
- (d) (7 points) Compute the marginal PDF  $f_Y(y)$ . Remember to specify the PDF at all values of  $y$ .
- (e) (7 points) Are  $X$  and  $Y$  independent? Justify your answer.
- (f) (5 points) Compute  $P(Y \leq \sqrt{X})$ .

3. (30 points.) Consider the following PDF:

$$f_{XY}(x, y) = \begin{cases} \frac{6}{19}(x^2 + y^3) & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (9 points.) Compute  $E(XY)$ ; this is called the *correlation* of  $X$  and  $Y$ .
- (b) (9 points.) Compute  $E(XY) - E(X)E(Y)$ ; this is called the *covariance* of  $X$  and  $Y$ .
- (c) (3 points.) Two random variables are called *uncorrelated* if their covariance is 0. Are  $X$  and  $Y$  uncorrelated?
- (d) (9 points.) Compute  $E(95(X(1 + Y) + 2Y(1 - X)))$  from your work in part (b) (i.e., don't do any new integration). Hint: if you did everything right in parts (a) and (b) this should be an integer.